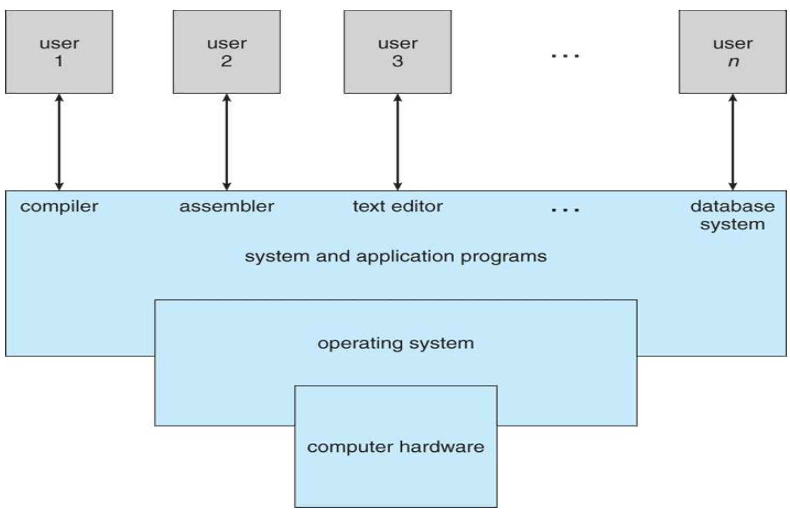
Unit-I. Introduction to Operating System, Process Management

Operating System:

- An operating system is system software that acts as interface between computer hardware and user.
- An operating system is a program which manages all the computer hardware's.
- It provides the base for application software's and other system software also.
- The operating system has two objectives such as:
  - Firstly, an operating system controls the computer's hardware.
  - The second objective is to provide an interactive interface to the user and interpret commands so that it can communicate with the hardware.
- The operating system is master software for almost every computer system. Without operating system, computer is nothing.
- It is a program that directly interacts with the hardware.
- The operating system is the first encoded with the Computer and it remains on the memory all time thereafter.
- Following diagram shows abstract view of operating system:



The abstract view divides the computer system into four parts. The first part is hardware (A bare machine without control). Above the bare machine, the operating system resides, which directly interact with the hardware or which makes the hardware to work. Third part is application program, user programs or any other system programs run above the operating system. Finally the user who interact through their program or through the application programs.

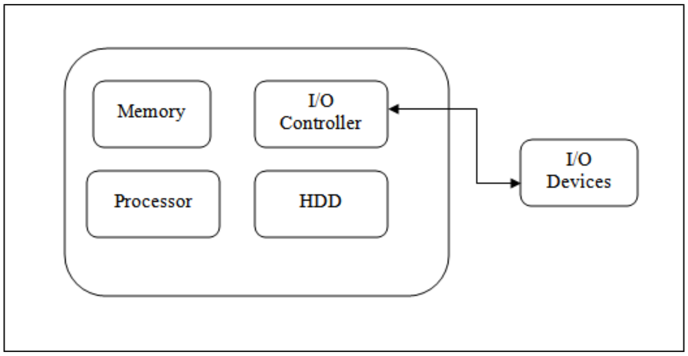
Operating System goals:

- The purpose of an operating system is to be provided an environment in which a user can execute programs.
- Its primary goals are to make the computer system convenience for the user.
- Its secondary goals are to use the computer hardware in efficient manner.

Task or operations of Operating system:

Following are several tasks performed by the operating system.

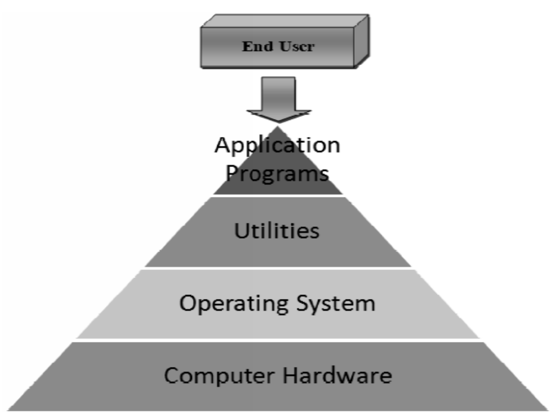
1) **Managing Hardware:**



- The prime objective of operating system is to manage & control the various hardware resources of a computer system.
- These hardware resources include processer, memory, and disk space and soon.
- The output result was display on monitor. In addition to communicating with the hardware the operating system provides an error handling procedure and display an error notification.
- If a device not functioning properly, the operating system cannot be communicate with the device.

2) **Providing an Interface:**

- It provides a stable and consistent way for applications to deal with the hardware without the user having known details of the hardware.
- The operating system organizes application so that users can easily access, use and store
- If the program is not functioning properly, the operating system again takes control, stops the application and displays the appropriate error message.
- **Computer system components are divided into 5 parts**
  - Computer hardware
  - Operating system
  - utilities
  - Application programs
  - End user



  - The Computer hardware: The central processing unit (CPU), the memory and I/O devices provides the basic computing resources for the system.
  - The operating System: It controls the hardware and coordinates its use among the various application programs for various users.
  - Utilities: Utility software helps to manage, maintain and control computer resources. Examples

of utility programs are antivirus software, backup software and disk tools.

- The application programs: Such as word processors, spreadsheets, compilers and Web browsers define the way in which these resources are used to solve users computing problem.
- End user: The person that uses the application software and demand for computer resources via. Application software towards operating system.

View of operating system

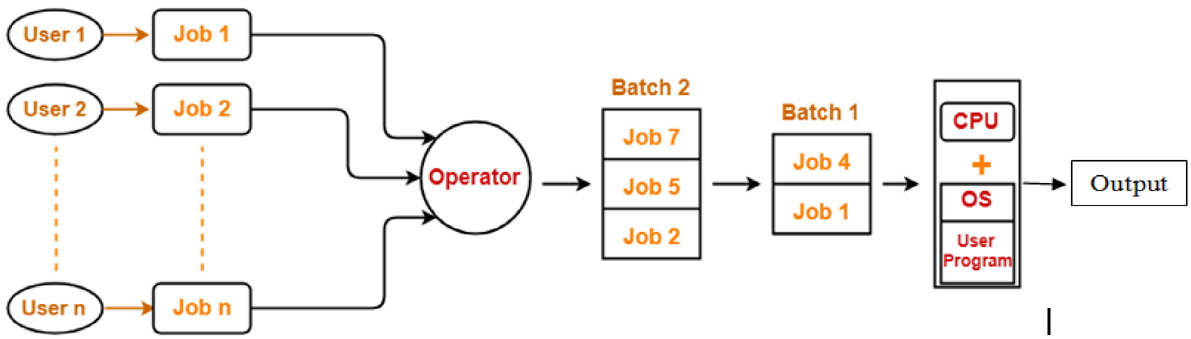
- **User view:** The user view of the computer varies by the interface being used. The examples are- windows XP, vista, windows7 etc. Most computer users it in front of personal computer (pc) in this case the operating system is designed mostly for easy use with some attention paid to resource utilization. Some user sit at a terminal connected to a mainframe/minicomputer. In this case other users are accessing the same computer through the other terminals. There user are share resources and may exchange the information. The operating system in this case is designed to maximize resources utilization to assume that all available CPU time, memory and I/O are use deficiently and no individual user takes more than his/her fair and share. The other users sit at workstations connected to network of other workstations and servers. These users have dedicated resources but they share resources such as networking and servers like file, compute and print server. Here the operating system is designed to compromise between individual usability and resource utilization.
- **System view:** From the computer point of view the operating system is the program which is most intermediate with the hardware. An operating system has resources as hardware and software which may be required to solve a problem like CPU time, memory space, file storage space and I/O devices and so on. That's why the operating system acts as manager of these resources. Another view of the operating system is it is a control program. A control program manages the execution of user programs to present the errors in proper use of the computer. It is especially concerned of the user the operation and controls the I/O devices.

Types of Operating System:

1) Batch Operating System/Early Operating System:

- Early computers (Since, 1960s') were physically large machine.
- The common input devices like punched cards, card readers, tape recorder were used to store the job (Program, input, Control instructions).
- First user has to prepare his/her job and store it on punched cards.
- The prepared jobs were handed over to computer operator of mainframe computer.
- The operator creates different batches of jobs depending upon properties job.
- Then operator loads these batches one after another towards CPU for execution and then output of jobs were stored on punched card.
- Here, loading (inputting) and unloading (outputting) of batches took more time and hence CPU remains ideal. And therefore, in batch operating <u>CPU utilization is low</u>.
- In these systems the user did not interact directly with the computer system therefore batch <u>operating system is non-interactive</u>.

Following diagram shows Batch operating system:



Advantages of Batch operating system-

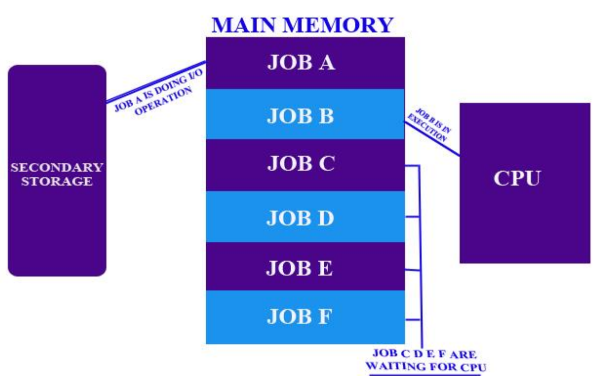
- It saves the time that was being wasted earlier for each individual process in context switching from one environment to another environment.
- No manual interface (intervention) is needed.

Disadvantages of Batch operating system-

- All the jobs of a batch are executed sequentially one after the other.
- The output is obtained only after all the jobs are executed. Thus, priority cannot be implemented if a certain job has to be executed on an urgent basis.
- The jobs of a particular batch might take long time for their execution. This might lead to starvation to other jobs in other batches.
- If the jobs of a batch require some I/O operation, then CPU must wait till the I/O operation gets completed.
- Since I/O devices are very slow, CPU remains idle for a long time. CPU cannot take any other job and execute it.

2) Multiprogramming operating system:

- In multiprogramming operating systems, several jobs are organized in memory in such a way that the CPU always has one job to execute that increases CPU utilization& this is main idea behind multiprogramming concept.
- The operating system keeps several jobs in memory simultaneously as shown in below figure.



Multiprogramming Operating System

- In above diagram, JOB A, JOB B, JOB C, JOB D, JOB E, JOB F are in memory. First, operating system picks JOB A and allocates CPU for execution.

- While processing of JOB A, other jobs were kept in waiting state. If JOB A wants some I/O operation then it transferred towards respective I/O device and immediately operating system picks next job JOB B for processing and allocates towards CPU. Now processing of JOB B is continues till it requires some I/O device or termination at the same time other remaining processes were kept in waiting state. Such process of processing the jobs continues one after another by the CPU till all were terminates or they require some I/O operation.
- In this process, CPU always has one job for processing. CPU does not remain ideal. Thus, in multi-programming operating system CPU utilization is increased.
- In this environment the operating system simply switches and executes another job.
- The multiprogramming operating system is sophisticated because the operating system makes decisions for the user. This is known as scheduling. If several jobs are ready to run at the same time the system choose one among them. This is known as CPU scheduling.

Advantage of the multi programmed system:

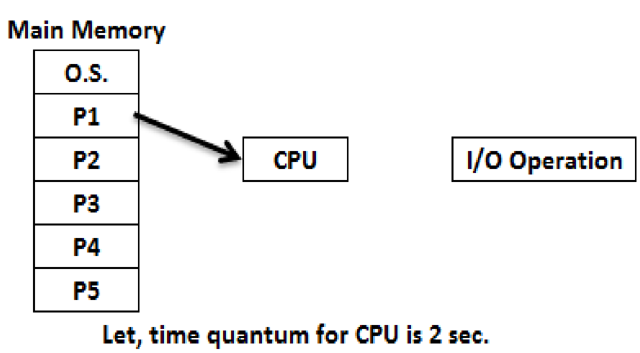
1. CPU always has one process or job for execution. Therefore it will not go in ideal state.
2. Multi-programmed operating system increases CPU utilization.

Disadvantage of the multi programmed system:

1. It does not provide user interaction with the computer system during the program execution.
2. If first job took more time for execution at that time remaining all jobs were kept in waiting state still they have less time for execution.

3) Multi-tasking (Time Sharing) operating System:

- Multi-tasking operating system is also known as time sharing operating system. It is extension of multiprogramming operating system.
- In multi-tasking system, we have multiple jobs or multiple processes in main memory & these processes are processed by CPU depending upon time quantum (time slice) defined by operating system.
- In this operating system, we can forcefully remove the process which is processing by the CPU depending upon time quantum of CPU.
- In this case, the CPU time is shared by different processes therefore it is called time sharing system.
- In this system, specific time slot is allocated for every process or program for execution.



Time-sharing Operating System

In above fig. of time sharing operating system, several processes P1, P2, P3, P4 and P5 are in main memory. Each process has assigned time slot for execution and it is shown in following table-

Process	Execution Time Slot
P1	10 sec
P2	4 sec
P3	5 sec
P4	2 sec
P5	6 sec

- Let, Operating system allocates time quantum of 2 sec. for CPU.

The processing of all processes is happen in following manner-

- The process P1 is allocating for processing by operating system towards CPU. CPU process P1 for 2 sec. only (Since, CPU has allocated 2 sec. time quantum for processing by O.S) after 2 sec. P1 process forcefully exit from CPU by O.S. Here, P1 completes its execution for 2 sec. only but it requires total 10 sec. to complete its execution therefore O.S. keep P1 in waiting queue and immediately process P2 allocated to CPU for processing. Here, also CPU process P2 for 2 sec. only, after 2 sec. P2 forcefully exit from CPU by O.S. and keep it in waiting queue. (Since, P2 requires total 4 sec. for to complete its execution).
- The above mentioned task is follow with every process by O.S. till it gets terminated (Completes its execution by given time slot) or it gets I/O instruction.
- Remember, in this operating system also CPU can execute only one process at a time depending upon time quantum.
- In this O.S., it seems multiple jobs are run simultaneously by CPU but it is not true. Because, CPU can allocate one process at one time & each process is then switching and process by CPU depending on time quantum allocated by operating system.
- CPU allocate process for execution, if execution time of process is not over then it is inserted in waiting queue & CPU allocate next process for processing. The next process processed depending upon time quantum if it's also execution time is not over then it will inserted in waiting queue. Such task is happen again and again till every processes terminates. Thus, In this O.S. **context switching** between process is done.

Advantages of time sharing operating system:

1) It provides the advantage of quick response.
2) This type of operating system avoids duplication of software.
3) It reduces CPU idle time.
4) In time sharing systems all the tasks are given specific time and task switching time is very less so applications don't get interrupted.
5) It seems, many applications can run at the same time.
6) Time sharing systems is better way to run a business having lots of tasks to be done and no task gets interrupted by the system.

Disadvantages of time sharing operating system:

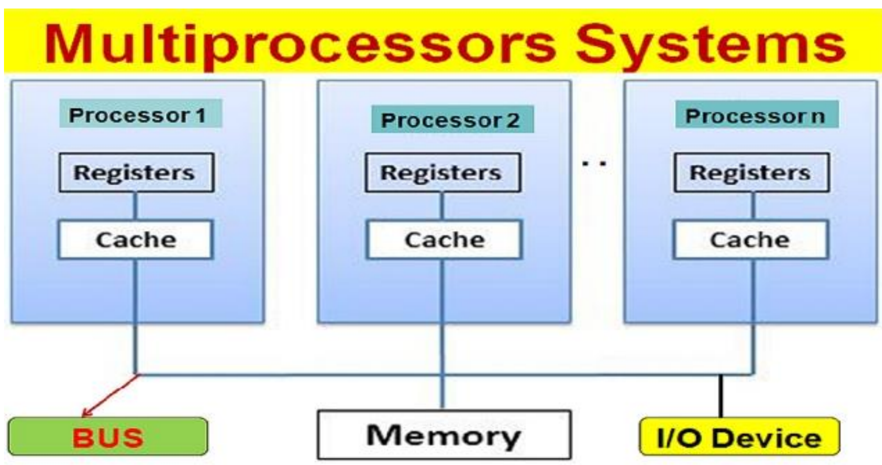
1) It is more complex than multiprogrammed operating system
2) The system must have memory management & protection, since several jobs are kept in memory at the same time.

3) Time sharing system must also provide a file system, so disk management is required.
4) It provides mechanism for concurrent execution which requires complex CPU scheduling.

4) Multi-processing Operating Systems/Parallel Systems/Tightly coupled Systems:

- These operating systems have multiple processors (more than one processor) which are used to process the programs or process.
- As the name implies there are multiple processors therefore multiple processes can be executed parallel by multiple processors. But each processor can process single process at a time.
- In this operating system all processors share the computer bus, clock, memory & peripheral devices commonly.
- We heard about dual core (two CPU's), quad core (four CPU's), octa core (Eight CPU's), hexacore (Sixteen CPU's) system. Depending upon number of CPU's, the system process different processes simultaneously.
- Ex: UNIX, LINUX

Following diagram shows multi-processing operating system:



Advantages of multi-processing operating system:

1) In this system, a functions can be properly distributed among several processors, then the failure of one process or will not halt (stop) the system, but slow it down.
2) Multiprocessor systems can share peripherals like storage, I/O devices, power supplies etc. and hence these systems are in economical scale.
3) In multi-processor system, number of processes computed/calculated per unit time. By increasing the number of processors move work can be done in less time. The speed up ratio with N processors is not N, but it is less than N. Because a certain amount of overhead is acquired in keeping all the parts working correctly.

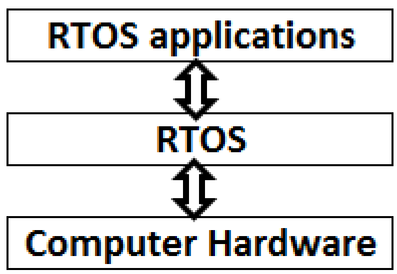
Disadvantages of multiprocessing systems

1) There are multiple processors in a multiprocessor system that share peripherals, memory etc. So, it is much more complicated to schedule processes. Hence, a more complex and complicated operating system is required in multiprocessor systems.
2) All the processors in the multiprocessor system share the memory. So a much larger pool of memory is required as compared to single processor systems.

5) Real time Operating Systems (RTOS):

- In RTOS, OS allocates process to CPU with rigid time bound (Deadline). If CPU unable to process the job in given time bound then entire RIOS failed. That is, overall work of RIOS is depends on exact time constraint.
- A real-time operating system (RTOS) is an operating system (OS) intended to serve real-time applications that process data depending on time bound, typically without buffer delays.
- A real-time system is a time-bound system which has well-defined, fixed time constraints. Processing must be done within the defined time constraints or the system will fail.
- Real time operating systems (RTOS) are used in such environments where a large number of events, mostly external to the computer system, must be accepted and processed in a short time or within certain deadlines.
- RIOS some applications are missile launching system, satellite launching system, flight simulation system, Air bag System (ABS) uses RTOS.

Following diagram Shows RTOS:



The real time operating systems can be of 2 types –

1) Hard Real Time operating system:

- These operating systems guarantee that critical tasks be completed within a range of time.
- Such type of RTOS, does not allow time delay to process the process.
- If delay occurs, hard RTOS failed immediately.
- For example,

1) Air bag system mounted in branded car uses hard RTOS.
2) A robot is hired to weld a car body, if robot welds too early or too late, the car cannot be sold, so it is a hard real time system that requires to complete car welding by robot hardly on the time.

2) Soft real time operating system:

- This operating system provides some relaxation in time limit or time bound i.e. soft RTOS allows some delay to process the process.
- It is a less restrictive type of real time system where a critical task gets priority over other tasks and retains that priority until it computes.
- For example, Multimedia systems, digital audio system etc. Programmer- defined and controlled processes are encountered in soft real time systems.

Advantages of real time operating systems

1) RTOS has maximum utilization of devices and system. Thus we get more output from all the resources.
2) Time assigned for shifting tasks in these systems are very less.
3) Focus on running applications and less importance to applications which are in queue.
4) Size of programs in RTOS are small, therefore RTOS can easily be embedded on systems like in transport and others RTOS application.

5) These types of systems are error free.
6) Memory allocation is best managed in these types of systems.

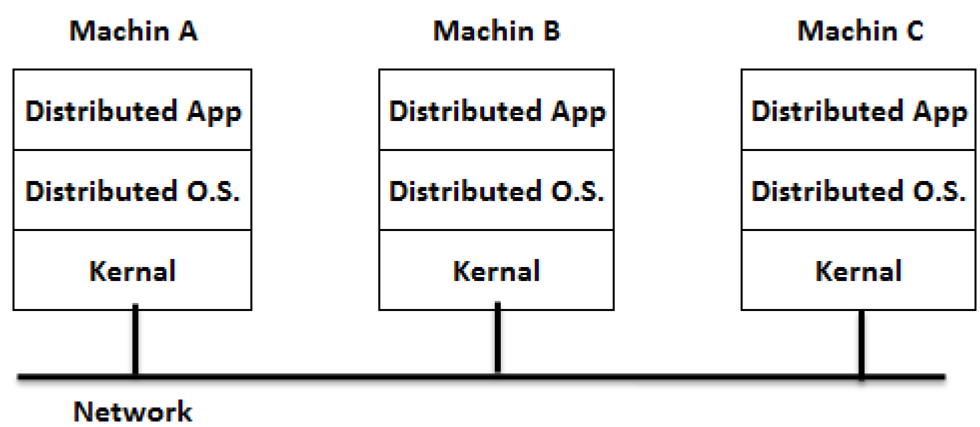
Disadvantages of real time system

1) A real time system is considered to function correctly only if it returns the correct result within the time constraints.
2) In RTOS, secondary storage is limited or missing whereas data is usually stored in short term memory RAM or ROM.

6) Distributed Operating System/Loosely Coupled Systems:

- The functionality of distributed systems depends on networking.
- In distributed operating system, the processors communicate with each other by various communication lines such as high speed buses or telephone lines.
- In contrast to tightly coupled systems, the processors in distributed operating system do not share memory or a clock. Instead, each processor has its own local memory.
- Distributed systems are able to share computational tasks and provide a rich set of features to the users.
- The main advantages of distributed systems are resource sharing, computation speed is high and reliability via communication network.

Following diagram shows distributed O.S.



Advantages of Distributed Operating System:

1) Failure of one system will not affect the other network communication, as all systems are independent from each other.
2) Electronic mail increases the data exchange speed.
3) Since resources are being shared, computation is highly fast and durable.
4) Load on host computer reduces.

Disadvantages of Distributed Operating System:

1) Security problem due to sharing.
2) Some messages can be lost in the network system.
3) Bandwidth is another problem if there is large data then all network wires to be replaced which tends to become expensive.
4) Overloading is another problem in distributed operating systems.

Operating System Services (Basic Functions of Operating System):

An Operating System provides services to both the users and to the programs.

- It provides an environment for programs to execute.
- It provides services to users to execute the programs in a convenient manner.

Following are a few common services provided by an operating system −

1) Program execution.
2) I/O operations management.
3) File System manipulation.
4) Communication/Networking
5) Error Detection.
6) Resource Allocation.
7) Protection.

Let's see all these services in details.

1) Program execution:

- Program execution is basic service provided by operating system.
- To execute the program, an operating system load many kinds of activities into the memory and run it. The program must be able to end its execution, either normally or abnormally.
- A process is basically a program in execution. A process includes the complete execution of the written program or code. There are some of the activities which are performed by the operating system in program execution:
  - ➢ The operating system Loads program into memory.
  - ➢ It also executes the program.
  - ➢ It handles the program's execution.
  - ➢ It provides a mechanism for process management
  - ➢ It provides a mechanism for process communication.

2) I/O Operation management:

- An I/O system comprises (includes) of I/O devices and their corresponding driver software.
- Drivers hide the particularities of specific hardware devices from the users.
- An Operating System manages the communication between user and device drivers.
  - ➢ I/O operation means read or write operation with any file or any specific I/O device.
  - ➢ Operating system provides the access to the required I/O device when required.

3) File system manipulation:

- A file represents a collection of related information which is permanently stored on secondary storage devices.
- Computers can store files on the disk (secondary storage), for long-term storage purpose. Examples of storage media include magnetic tape, magnetic disk and optical disk drives like CD, DVD.
- Each of these media has its own properties like speed, capacity, data transfer rate and data access methods.
- A file system is normally organized into directories for easy navigation and usage. These

directories may contain files and other directions.

Following are the major activities of an operating system with respect to file management −

- Program needs to read a file or write a file.
- The operating system gives the permission to the program for operation on file.
- Permission varies from read-only, read-write, denied and so on.
- Operating System provides an interface to the user to create/delete files.
- Operating System provides an interface to the user to create/delete directories.
- Operating System provides an interface to create the backup of file system.

4) Communication / Networking:

- We know that, the distributed systems uses number of processors located in different geographical area and these processor do not share memory, peripheral devices, or a clock. Every individual processor has its local memory, clock etc.
- The operating system manages communications between all the processes that happen between different systems in the network.
- Multiple processes communicate with one another through communication lines in the network.
- The OS handles routing and connection strategies, and the problems of connection and security.
- Following are the major activities of an operating system with respect to communication −
  - Two processes often require data to be transferred between them.
  - Both the processes can be on one computer or on different computers, but are connected through a computer network.
  - Communication may be implemented by two methods, either by Shared Memory or by Message Passing.

5) Error handling:

- Errors can occur anytime and anywhere.
- An error may occur in CPU, in I/O devices or in the memory hardware.
- Following are the major activities of an operating system with respect to error handling −
  - The OS constantly checks for possible errors.
  - The OS takes an appropriate action against the happened error and ensure correct and consistent computing.

6) Resource Management:

- In case of multi-user or multi-tasking environment, resources such as main memory, CPU and files storage etc. are to be allocated to each user this is done by operating system.
- Following are the major activities of an operating system with respect to resource management −
  - The OS manages all kinds of resources using schedulers.
  - CPU scheduling algorithms are used for better utilization of CPU.

7) Protection:

- Protection refers to a mechanism or a way to control the access of programs, processes, or users to the resources defined by a computer system with the help of operating system.
- We know that, in computer system multiple users and concurrent execution of multiple processes are happen, such various processes must be protected from each other's activities and such

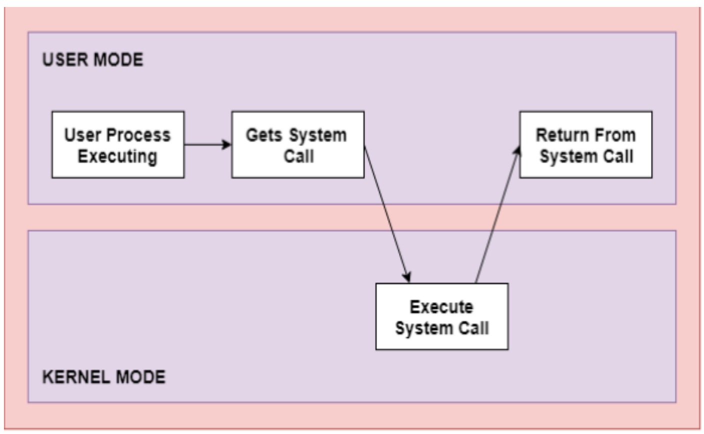
protection is provided by operating system.

Following are the major activities of an operating system with respect to protection −

- The OS ensures that all access to system resources is controlled.
- The OS ensures that external I/O devices are protected from invalid access attempts.
- The OS provides authentication features for each user by means of passwords.

System Calls:

- The interface between a process and an operating system is provided by system calls.
- In general, system calls are available in assembly language instructions.
- System calls are usually made when a process is in user mode and requires access to a resource. Then it requests the kernel to provide the resource via a system call.
- They are also included in the manuals used by the assembly level programmers.
- Some systems allow system calls to be made directly from a high level language program like C, BCPL and PERL etc.
- Following figure shows execution of system call:



The above diagram shows execution of system call:

- The processes execute normally in the user mode until a system call interrupts this.
- Then the system call is executed on a priority basis in the kernel mode.
- After the execution of the system call, the control returns to the user mode and execution of user processes can be resumed.

In general, system calls are required in the following situations −

- If a file system requires the creation or deletion of files.
- Reading and writing from files also require a system call.
- Creation and management of new processes.
- Network connections also require system calls. This includes sending and receiving packets.
- Access to a hardware devices such as a printer, scanner etc. requires a system call.

Types of System Calls:

There are mainly five types of system calls. These are explained in detail as follows −

1) Process Control:

- These system calls deals with processes such as process creation, process termination etc.

2) File Management:

- These system calls are responsible for file manipulation such as creating a file, reading a file, writing into a file etc.

3) Device Management:

- These system calls are responsible for device manipulation such as reading from device buffers, writing into device buffers etc.

4) Information Maintenance:

- These system calls handle information and its transfer between the operating system and the user program.

5) Communication:

- These system calls are useful for inter process communication. They also deal with creating and deleting a communication connection.

Some of the examples of all the above types of system calls in Windows and Unix are given as follows:

Types of System Calls	Windows	Linux / Unix
Process Control	CreateProcess()	fork()
	ExitProcess()	exit()
	WaitForSingleObject()	wait()
File Management	CreateFile()	open()
	ReadFile()	read()
	WriteFile()	write()
	CloseHandle()	close()
Device Management	SetConsoleMode()	ioctl()
	ReadConsole()	read()
	WriteConsole()	write()
Information Maintenance	GetCurrentProcessID()	getpid()
	SetTimer()	alarm()
	Sleep()	sleep()
Communication	CreatePipe()	pipe()
	CreateFileMapping()	shmget()
	MapViewOfFile()	mmap()

Process Management

Process:

- A process or task is a program in execution.
- A process is mainly a program in execution where the execution of a process must progress in sequential order or based on some priority or algorithms.
- When a program gets loaded into the memory, it is said to as a process.
- In other words, process is an entity that represents the fundamental working that has been assigned to a system.
- The OS helps you to create, schedule, and terminates the processes which is used by CPU.
- A process created by the main process is called a child process.
- Process operations can be easily controlled with the help of PCB (Process Control Block). You can consider it as the brain of the process, which contains all the crucial information related to processing like process id, priority, state, CPU registers, etc.
- For example, when we write a program in C or C++ and compile it, the compiler creates binary code. The original code (source code) and binary code are both programs. When we actually run the binary code, it becomes a process.

Difference between program & process:

Following points shows difference between program and process:

Program:

- A program is a set of statements or instructions written in programing languages.
- A program by itself is not a process. It is just contents of a file stored on disk.
- A program is a passive entity.

Process:

- A program in execution is known as a process.
- Every Process has program counter specifying the next instruction to execute.
- Also, set of associated resources may be shared among several process with some scheduling algorithm being used to determinate when the stop work on one process and service a different one.
- Process is an active entity because it is in running state.

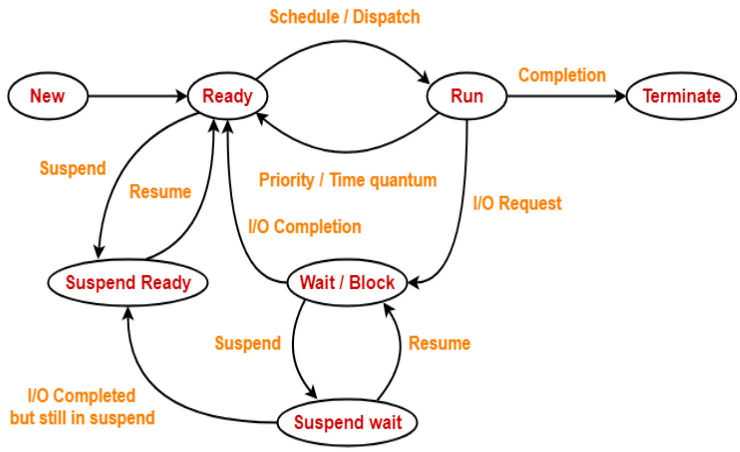
What is Process Management?

- Process management involves various tasks like creation, scheduling, termination of processes, and a dead lock.
- Process is a program that is under execution, which is an important part of modern-day operating systems.
- The OS must allocate resources that enable processes to share and exchange information. It also protects the resources of each process from other methods and allows synchronization or organization among processes.
- It is the job of OS to manage all the running processes of the system. It handles operations by performing tasks like process scheduling and such as resource allocation.

Process state (Process Life Cycle):

- A process state is a condition of the process at a specific instant of time. It also defines the current position of the process.

- As a process executes, it changes state. The state of a process is defined by the correct activity of that process.
- Many processes may be in ready and waiting state at the same time. But only one process can be running on any processor at any instant.
- Following diagram shows process states.



Process State Diagram

There are mainly seven stages or states of a process which are:

- **New**: The new process is created when a specific program calls from secondary memory/ hard disk to primary memory/ RAM.
- **Ready:** In a ready state, the process should be loaded into the primary memory, which is ready for execution.
- **Executing (Running):** The process is in execution state allocated to CPU.
- **Waiting**: The process is waiting for the allocation of CPU time and other resources for execution.
- **Blocked:** It is a time interval when a process is waiting for an event like I/O operations to complete.
- **Suspended:** Suspended state defines the time when a process is ready for execution but has not been placed in the ready queue by OS.
- **Terminated:** Terminated state specifies the time when a process is terminated or end.

Following table shows Process state with current memory:

State	Present in Memory
New state	Secondary Memory
Ready state	Main Memory
Run state	Main Memory
Wait state	Main Memory
Suspend wait state	Secondary Memory
Suspend ready state	Secondary Memory

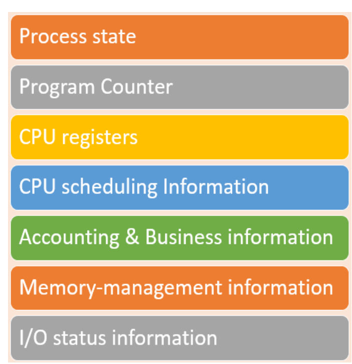
Terminate state	–

Points to remember:

- After completing every step, all the resources are used by a process, and memory becomes free.
- The minimum number of states through which a process compulsorily goes through is 4.
- These states are new state, ready state, run state and terminate state.
- However, if a process also requires the I/O operation, then minimum number of states is 5.

Process Control Block (PCB):

- The full form of PCB is Process Control Block. It is a data structure which is maintained by the Operating System for every process.
- Every process is represented in the operating system by a process control block, which is also called a task control block.
- Process operations can be easily controlled with the help of PCB (Process Control Block). You can consider it as the brain of the process, which contains all the crucial information related to processing like process id, priority, state, CPU registers, etc.
- The PCB should be identified by an integer value called Process ID (PID). It helps you to store all the information required to keep track of all the running processes.
- It is also responsible for storing the contents of processor registers. These are saved when the process moves from the running state and then returns back to it.
- The information is quickly updated in the PCB by the OS as soon as the process makes the state transition.
- PCB of each process resides in the main memory.
- There exists only one PCB corresponding to each process.
- PCB of all the processes are present in a linked list form.
- Following diagram shows PCB:



There are 7 important or major components of PCB which is discussed as follow:

1) **Process state:** A process can be new, ready, running, waiting, etc.
2) **Program counter:** The program counter stores the address of the next instruction, which should be executed for that process.
3) **CPU registers:** This component includes accumulators, index and general-purpose registers, and information of condition code.
4) **CPU scheduling information**: This component includes a process priority, pointers for scheduling queues, and various other scheduling parameters.
5) **Accounting and business information:** It includes the amount of CPU and time utilities

like real time used, job or process numbers, etc.

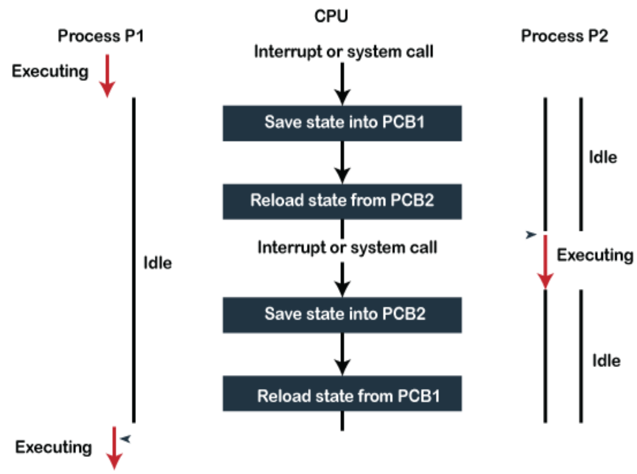
6) **Memory-management information:** This information includes the value of the base and limit registers, the page, or segment tables. This depends on the memory system, which is used by the operating system.

7) **I/O status information:** This block includes a list of open files, the list of I/O devices that are allocated to the process, etc.

Context Switching:

- Context Switching involves storing the context or state of a process so that it can be reloaded when required and execution can be resumed from the same point as earlier.
- Basically, it is a feature of a multitasking operating system and allows a single CPU to be shared by multiple processes.

Following diagram shows context switching:



In the above diagram, initially Process 1 is running. Process 1 is switched out and Process 2 is switched in because of an interrupt or a system call. Context switching involves saving the state of Process 1 into PCB1 and loading the state of process 2 from PCB2. After some time again a context switch occurs and Process 2 is switched out and Process 1 is switched in again. This involves saving the state of Process 2 into PCB2 and loading the state of process 1 from PCB1.

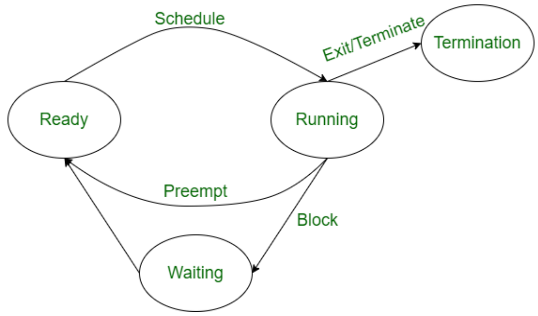
Context Switching Steps:

The steps involved in context switching are as follows −

- Save the state or context of the process that is currently running on the CPU. Update the process control block and other important fields.
- Move the process control block of the above process into the relevant queue such as the ready queue, I/O queue etc.
- Select a new process for execution.
- Update the process control block of the selected process. This includes updating the process state to running.
- Update the memory management data structures as required.
- Restore the state or context of the process that was previously running when it is loaded again on the processor. This is done by loading the previous values of the process from PCB and registers.

Operations on Process:

The execution of a process is a complex activity. It involves various operations. Following diagram shows operations that are performed while execution of a process:



From the above diagram there are different operations that are performed on a process while execution, which are as follow:

1) Creation.
2) Scheduling / Dispatching
3) Blocking
4) Preemption
5) Termination

Let's see all these operations in details-

1) Creation:

- This the initial step of process execution activity.
- Process creation means the construction of a new process for the execution.
- This might be performed by operating system, user or old process itself.
- There are several events that leads to the process creation. Some of the events are following:
  - When we start the computer, system creates several background processes.
  - A user may request to create a new process.
  - A process can create a new process itself while executing.

2) Scheduling/Dispatching:

- In this operation of process, the state of the process is changed from ready to running.
- It means the operating system puts the process from ready state into the running state.
- Dispatching is done by operating system when the resources are free or the process has higher priority than the ongoing process.

3) Blocking:

- When a process demands an input-output device then system call generates that blocks the process and operating system put that process in block mode.
- Block mode is basically a mode where process waits for input-output resource. Hence on the demand of process itself, operating system blocks the process and dispatches another process to the processor.
- In process blocking operation, the operating system puts the process in 'waiting' state.

4) Preemption:

- When a timeout occurs and the process had not been terminated in the allotted time interval and next process is ready to execute, then the operating system preempts (prevents) the process.
- This operation is only valid where CPU scheduling supports preemption.
- Basically this happens in priority scheduling where on the incoming of high priority process the ongoing process is preempted.
- In process preemption operation, the operating system puts the process in 'ready' state.

5) Termination:

- Process termination is the activity of ending the process.
- In other words, process termination is the relaxation of computer resources taken by the process

for the execution.

- Like creation, in termination also there may be several events that may lead to the process termination. Some of them are:
  - ➢ Process completes its execution fully and it indicates to the OS that it has finished.
  - ➢ Operating system itself terminates the process due to service errors.
  - ➢ There may be problem in hardware that terminates the process.
  - ➢ One process can be terminated by another process.

Co-Operating Process:

- In the computer system, there are many processes which may be either independent processes or cooperating processes that run in the operating system.

1) Independent Processes:

Two processes are said to be independent if the execution of one process does not affect the execution of another process.

- ➢ It is clear that any process which does not share any data (temporary or persistent) with any another process then the process independent.

2) Cooperative Processes:

- Two processes are said to be cooperative if the execution of one process affects the execution of another process. These processes need to be synchronized so that the order of execution can be guaranteed.
  - ➢ The cooperating process is one which shares data with another process.

Reasons to need cooperating processes: (Advantages of Cooperating Process)

There are many reasons for the requirement of cooperating processes. Some of these are given as follows,

I) Modularity:

- ➢ Modularity means dividing complicated tasks into smaller subtasks.
- ➢ These subtasks can completed by different cooperating processes. This leads to faster and more efficient completion of the required tasks.

II) Information Sharing:

- ➢ Sharing of information between multiple processes can be accomplished using cooperating processes.
- ➢ This may include access to the same files. A mechanism is required so that the processes can access the files in parallel to each other.

III) Convenience:

- ➢ There are many tasks that a user needs to do such as compiling, printing, editing etc. These task becomes convenient if they can be managed by cooperating processes.

IV) Computation Speedup:

- ➢ Subtasks of a single task can be performed parallel using cooperating processes. This increases the computation speedup as the task can be executed faster. However, this is only possible if the system has multiple processing elements.